

Assignment 2 (Sep. 21-28)

Problem 1:

An art thief has broken into a museum and wants to steal as many art pieces as they can. However, they are on their own and can only carry a limited amount of weight. They want to steal art totaling as much value as they can. The following is a collection of art pieces that they can steal and their weights and their values.

Piece	1	2	3	4	5	6
Weight (kg)	10	15	7	8	5	3
Value (\$)	50	100	30	50	12	15

- (a) Formulate an integer program that models the thief's situation, assuming that they can only hold 30kg of art. Clearly explain your variables, objective function and constraints. **Solution:** For each art piece, the thief needs to decide whether or not to take that art piece. Let x_i be a variable which we will imagine is either 0 or 1, which is 1 if the art piece is taken and 0 if it is not.

The objective is to maximize the total value of the art, i.e.

$$50x_1 + 100x_2 + 30x_3 + 50x_4 + 12x_5 + 15x_6.$$

The constraint is that the weight of the art is at most 30kg:

$$10x_1 + 15x_2 + 7x_3 + 8x_4 + 5x_5 + 3x_6 \leq 30.$$

finally, each x_i must be between 0 and 1 and an integer, so that

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0,$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \leq 1,$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \in \mathbb{Z}.$$

Thus, the linear program is

$$\begin{aligned} \max \quad & 50x_1 + 100x_2 + 30x_3 + 50x_4 + 12x_5 + 15x_6 \\ \text{such that} \quad & 10x_1 + 15x_2 + 7x_3 + 8x_4 + 5x_5 + 3x_6 \leq 30 \\ & x_1, x_2, x_3, x_4, x_5, x_6 \geq 0 \\ & x_1, x_2, x_3, x_4, x_5, x_6 \leq 1 \\ & x_1, x_2, x_3, x_4, x_5, x_6 \in \mathbb{Z}. \end{aligned}$$

- (b) Suppose that in fact, the art thief has two bags, each of which can hold 15 kg of art. How do we need to change the previous integer program to model this change?

Solution: For each art piece, the thief not only needs to decide whether or not to take that art piece, but also which bag to put it in. For each art piece, let x_{i1} and x_{i2} be two

variables. We will imagine that x_{i1} is 1 if art piece i is placed in bag 1, and 0 otherwise. Similarly, we will imagine that x_{i2} is 1 if art piece i is placed in bag 2, and 0 otherwise.

The objective is to maximize the total value of the art, and it doesn't matter which bag we put it each piece in, so our objective is to maximize

$$50(x_{11} + x_{12}) + 100(x_{21} + x_{22}) + 30(x_{31} + x_{32}) + 50(x_{41} + x_{42}) + 12(x_{51} + x_{52}) + 15(x_{61} + x_{62}).$$

The constraint is that the weight of the art in each bag is at most 15.

$$10x_{11} + 15x_{21} + 7x_{31} + 8x_{41} + 5x_{51} + 3x_{61} \leq 15.$$

$$10x_{12} + 15x_{22} + 7x_{32} + 8x_{42} + 5x_{52} + 3x_{62} \leq 15.$$

We must also place each art piece in at most 1 bag, so that

$$x_{11} + x_{12}, x_{21} + x_{22}, x_{31} + x_{32}, x_{41} + x_{42}, x_{51} + x_{52}, x_{61} + x_{62} \leq 1.$$

Finally, each x_{ij} must be between 0 and 1 and an integer, so that

$$x_{11}, x_{21}, x_{31}, x_{41}, x_{51}, x_{61}, x_{12}, x_{22}, x_{32}, x_{42}, x_{52}, x_{62} \geq 0,$$

$$x_{11}, x_{21}, x_{31}, x_{41}, x_{51}, x_{61}, x_{12}, x_{22}, x_{32}, x_{42}, x_{52}, x_{62} \leq 1.$$

$$x_{11}, x_{21}, x_{31}, x_{41}, x_{51}, x_{61}, x_{12}, x_{22}, x_{32}, x_{42}, x_{52}, x_{62} \in \mathbb{Z}.$$

Thus, the linear program is

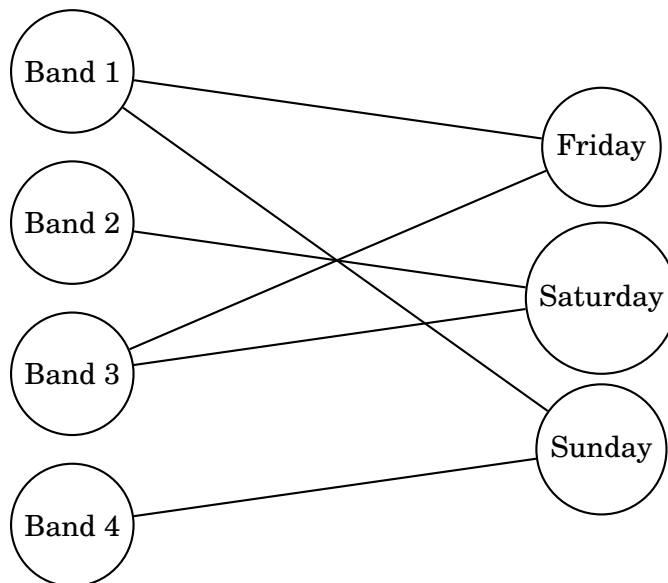
$$\begin{aligned} \max \quad & 50(x_{11} + x_{12}) + 100(x_{21} + x_{22}) + 30(x_{31} + x_{32}) + 50(x_{41} + x_{42}) + \\ & 12(x_{51} + x_{52}) + 15(x_{61} + x_{62}) \\ \text{such that} \quad & 10x_{11} + 15x_{21} + 7x_{31} + 8x_{41} + 5x_{51} + 3x_{61} \leq 15 \\ & 10x_{12} + 15x_{22} + 7x_{32} + 8x_{42} + 5x_{52} + 3x_{62} \leq 15 \\ & x_{11}, x_{21}, x_{31}, x_{41}, x_{51}, x_{61}, x_{12}, x_{22}, x_{32}, x_{42}, x_{52}, x_{62} \geq 0 \\ & x_{11}, x_{21}, x_{31}, x_{41}, x_{51}, x_{61}, x_{12}, x_{22}, x_{32}, x_{42}, x_{52}, x_{62} \leq 1 \\ & x_{11}, x_{21}, x_{31}, x_{41}, x_{51}, x_{61}, x_{12}, x_{22}, x_{32}, x_{42}, x_{52}, x_{62} \in \mathbb{Z} \end{aligned}$$

Problem 2:

A music venue needs to schedule acts for the week. They operate Friday through Sunday, and each day, they schedule one act. There are 4 possible bands that they can schedule, but each band has only certain nights they can play. Below is a table that shows which bands are available on which nights.

	Friday	Saturday	Sunday
Band 1	✓		✓
Band 2		✓	
Band 3	✓	✓	
Band 4			✓

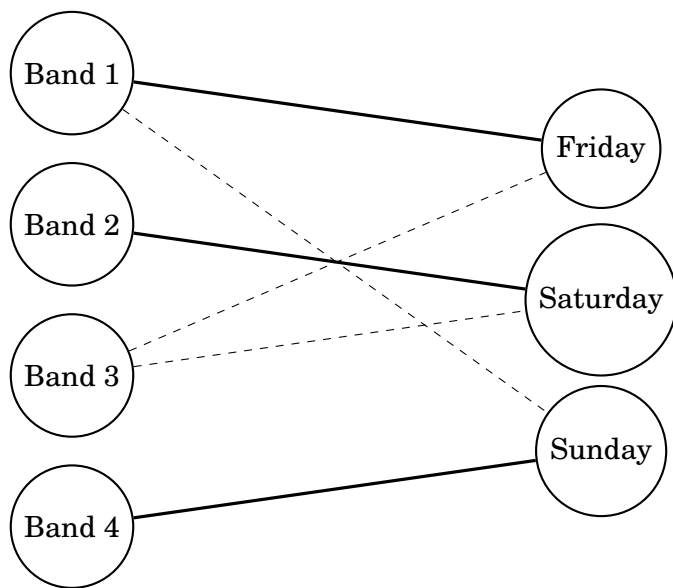
- (a) Draw the availability data as a bipartite graph. (A graph $G = (V, E)$ is bipartite if V can be partitioned into two parts A and B such that every edge e in E has exactly one end in A and one end in B .)



Solution:

- (b) What does a valid scheduling correspond to in this bipartite graph? Find and draw a valid schedule for this music venue.

Solution: A scheduling corresponds to a maximum matching with all of the weekdays matched.



- (c) Write down an integer program that solves this problem. The integer program should be of the form

$$\begin{aligned}
 & \min \quad c^T x \\
 & \text{such that } Ax \leq b \\
 & \quad 1 \geq x \geq 0 \\
 & \quad x \in \mathbb{Z}^k,
 \end{aligned}$$

where A is a matrix and c is a vector, and k is an integer.

Solution: x should be a vector with 6 entries; one for every edge. For each edge $\{i, j\}$, we want x_{ij} to be 1 if that edge is chosen in the matching and 0 if it isn't.

The objective is then just the number of edges chosen, which is given by $\sum_{\{i,j\} \in E} x_{ij}$, so $c = (1, 1, 1, 1, 1, 1)$.

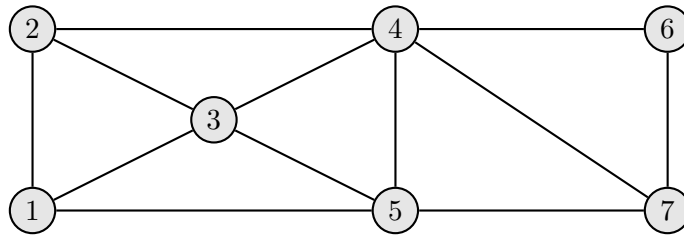
There should be one constraint per vertex; for a vertex i , each constraint should be of the form $\sum_{j: \{i,j\} \in E} x_{ij} \leq 1$. Since there are 7 vertices and 6 edges, the matrix A should be

Vertex/Edge	{1,F}	{1,Su}	{2,Sa}	{3, F}	{3, Sa}	{4,Su}
Band 1	1	1	0	0	0	0
Band 2	0	0	1	0	0	0
Band 3	0	0	0	1	1	0
Band 4	0	0	0	0	0	1
Friday	1	0	0	1	0	0
Saturday	0	0	1	0	1	0
Sunday	0	1	1	0	0	1

Problem 3:

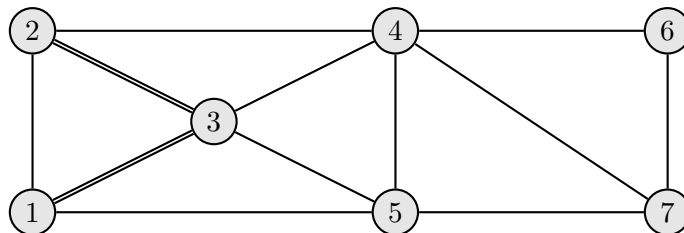
We are given a graph G representing a network. A *triangle* in the graph is a set of three distinct nodes i, j, k such that all pairs ij, jk, ik are edges in G . Our goal is to pick as many triangles as possible so that no pair of triangles share an edge. Note that triangles are allowed to share nodes (triangles represent self-contained subnetworks that cannot share communication lines with other subnetworks). For the graph below the set of triangles $\{123, 345, 467\}$ would be a feasible solution; the set of triangles $\{123, 135\}$ would not be a feasible solution (the edge 13 is used by two triangles).

- (a) Formulate the above-mentioned problem as an IP for the following graph G .



Hint: First, list all the triangles in G . Then introduce a binary variable for each triangle. Then, for each edge ij in the graph, write a constraint that “says:” among the triangles which use this edge ij we may pick at most one of them.

- (b) Now, suppose we are given an arbitrary graph G with node set V and edge set E . Let T denote the set of triangles in G . Write down your IP model for this general set-up.
- (c) Next consider a graph G whose edge set E is made up from two types of edges E_1 and E_2 . In these graphs the edges in E_1 have the same rule as before; but, the edges in E_2 can be used by at most two triangles (instead of at most one). For the graph below where $E_2 := \{13, 23\}$, formulate the new problem as an IP. Your formulation should be easily generalizable to any graph G .



Solution:

Given a graph G with vertex set V and edge set E , we denote by T the set of all distinct triangles ijk in G . Without loss of generality, we can and do assume that $(\forall (i, j, k) \in T) i < j < k$. So, we represent the triangles as ordered triples of vertices.

((a)) Let T denote the set of triangles for the given graph G in part (a). Therefore,

$$T = \{(1, 2, 3), (2, 3, 4), (1, 3, 5), (3, 4, 5), (4, 5, 7), (4, 6, 7)\}.$$

For each triangle $(i, j, k) \in T$ with vertices $\{i, j, k\} \subseteq \{1, 2, \dots, 7\}$, we define the binary variable $y_{i,j,k}$ which takes the value 1 if the triangle is picked in the selection, and 0 otherwise. Our IP will take the form

$$\begin{aligned} \max \quad & y_{1,2,3} + y_{3,4,5} + y_{1,3,5} + y_{2,3,4} + y_{4,5,7} + y_{4,6,7} \\ \text{subject to} \quad & y_{1,2,3} + y_{2,3,4} \leq 1 && \text{(edge 23)} \\ & y_{1,2,3} + y_{1,3,5} \leq 1 && \text{(edge 13)} \\ & y_{3,4,5} + y_{1,3,5} \leq 1 && \text{(edge 35)} \\ & y_{2,3,4} + y_{3,4,5} \leq 1 && \text{(edge 34)} \\ & y_{3,4,5} + y_{4,5,7} \leq 1 && \text{(edge 45)} \\ & y_{4,5,7} + y_{4,6,7} \leq 1 && \text{(edge 47)} \\ & 0 \leq y_{i,j,k} \leq 1, \quad y_{i,j,k} \text{ is an integer, } \forall (i, j, k) \in T. \end{aligned}$$

The last line of constraints ensure $y_{i,j,k}$ variables are all binary values, as desired. The remaining constraints guarantee that for each edge in the graph we pick at most one triangle which uses that edge. The objective function counts the number of triangles picked which is maximized as desired.

Note that we did not write down explicitly the constraints on the edges 12, 15, 24, 46, 57, 67 since for each of these edges there is at most one triangle which may use it (making the underlying constraint redundant). It is acceptable to include these constraints as part of your answer for this question.

((b)) Let T denote the set of triangles for the given graph G . For each triangle $(i, j, k) \in T$ with vertices $\{i, j, k\} \subseteq V$, we define the binary variable $y_{i,j,k}$ which takes the value 1 if the triangle is picked in the selection, and 0 otherwise. We assume without loss of generality that the edges in E are recorded as ab with $a < b$. Our IP will take the form

$$\begin{aligned} \max \quad & \sum_{(i,j,k) \in T} y_{i,j,k} \\ \text{subject to} \quad & \sum (y_{i,j,k} : (i, j, k) \in T \text{ and } \{a, b\} \subset \{i, j, k\}) \leq 1 && \forall ab \in E, \\ & 0 \leq y_{i,j,k} \leq 1, \quad y_{i,j,k} \text{ is an integer, } \forall (i, j, k) \in T. \end{aligned}$$

The last line of constraints ensure $y_{i,j,k}$ variables are all binary values, as desired. The remaining constraints guarantee that for each edge in the graph we pick at most one triangle

which uses that edge. Notice, the sum in the LHS includes exactly the $y_{i,j,k}$ variables for the triangles which include that fixed edge (for that constraint) ab . The objective function counts the number of triangles picked which is maximized as desired.

This time we did write down explicitly the constraints for every edge $ab \in E$.

((c)) Let T denote the set of triangles for the given graph G . For each triangle $(i, j, k) \in T$ with vertices $\{i, j, k\} \subseteq V$, we define the binary variable $y_{i,j,k}$ which takes the value 1 if the triangle is picked in the selection, and 0 otherwise. We assume without loss of generality that the edges in E are recorded as ab with $a < b$. Our IP will take the form

$$\begin{aligned}
& \max \quad \sum_{(i,j,k) \in T} y_{i,j,k} \\
\text{subject to} \quad & \sum (y_{i,j,k} : (i, j, k) \in T \text{ and } \{a, b\} \subset \{i, j, k\}) \leq 1 & \forall ab \in E_1, \\
& \sum (y_{i,j,k} : (i, j, k) \in T \text{ and } \{a, b\} \subset \{i, j, k\}) \leq 2 & \forall ab \in E_2, \\
& 0 \leq y_{i,j,k} \leq 1, \quad y_{i,j,k} \text{ is an integer, } \forall (i, j, k) \in T.
\end{aligned}$$

The last line of constraints ensure $y_{i,j,k}$ variables are all binary values, as desired. The first group of constraints guarantee that for each edge in E_1 , we pick at most one triangle which uses that edge. The remaining constraints guarantee that for each edge in E_2 , we pick at most two triangles which use that edge. The objective function counts the number of triangles picked which is maximized as desired.